Do Cointegration-based Pairs Remain Profitable?

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October 2026

Abstract

This study examines whether cointegration-based pairs remain profitable out of sample across estimators, stock groupings and window lengths, and whether the mean reversion found at formation persists. Spreads of S&P 500 constituents are formed by 5 estimators of the cointegrating vector within sectors and correlation-based clusters over 1- to 5-year formation windows, and are traded by a 2-standard-deviation threshold rule. We compare formation statistics, out-of-sample mean reversion and net returns with their values on spreads formed from independent random walks. No formation choice makes the spreads profitable on its own. In the testing sample from 2020 to 2025, the spreads with the smallest formation p-values lose 0.7% to 1.3% a year, and their mean reversion persists little more than that of random-walk spreads. The result agrees with the decline in pairs trading profits reported since 2010.[1]

1 Introduction

A pairs trade takes offsetting positions in assets whose prices have deviated from a long-run equilibrium, and closes them when the deviation reverts. The strategy has two stages, a formation stage that selects the assets and estimates the spread, and a trading stage that opens positions on divergence and closes them on mean reversion. Cointegration is a standard formation tool, as it identifies stationary linear combinations of nonstationary prices [1, 2]. The framework extends to baskets through the Johansen procedure [3] and principal components (PCs) of price levels [4], and to groups of assets defined by clustering [5]. Do cointegration-based pairs remain profitable out of sample?

The empirical literature documents a decline in profitability. Clegg and Krauss [6] report a net return of −2.76% a year from 2010 to 2015 for partial cointegration, and Bui and Ślepaczuk [7] find only 0.06% a year for Engle–Granger pairs on the Nasdaq 100 from 2002 to 2018. Reichmann [8] reports a cumulative net loss of 5.49% for pairs selected by price distance from 2020 to 2023. Cointegration found at formation also rarely persists: the share of pairs that remain cointegrated a year later is about the size of the test [9], and Farago and Hjalmarsson [10] demonstrate that truly cointegrated stocks would imply Sharpe ratios above 10, far beyond those observed.

This study applies 5 estimators of the cointegrating vector within sectors and correlation-based clusters of S&P 500 constituents, over formation windows of 1 to 5 years, and trades the spreads from 2010 to 2025 by the threshold rule of Gatev et al. [11] over horizons of 1 to 12 months. We also examine whether the mean reversion found at formation persists, comparing every statistic with its value on spreads formed from independent random walks. No formation choice makes the spreads profitable on its own. From 2020 to 2025, the top decile of every method loses 0.70% to 1.34% a year, the formation p-value does not consistently rank returns, and neither the grouping nor the window changes the sign of the losses. This mean reversion persists little more than that of spreads formed on random walks.

The remainder of the report is organized as follows. Section 2 describes the data. Section 3 states the formation, the random walk null and the trading rule. Section 4 reports the results, and Section 5 concludes.

[1] Code and replication data are available at *GitHub*.

2 Data

The sample uses the point-in-time S&P 500 membership [12], so that stocks enter and leave with the index through mergers, spin-offs and market-cap changes. Daily adjusted closing prices from 2005 to 2025 and the sector of each stock are from Yahoo Finance, which holds no prices for stocks delisted before 2026. A stock is therefore eligible at a formation date if it is an index member on that date, still trades at the end of 2025 and has a complete price record over the formation window, which keeps 330 of the 499 members in January 2010 and 447 of the 505 in January 2020. Formation dates are the first trading days of each month. The training sample comprises the trading windows that start between 2010 and 2019, and the testing sample those that start between 2020 and 2025.

3 Methodology

3.1 Spreads and Grouping

Write $P_{i,t}$ for the adjusted closing price of stock i on day t. The formation window Φ covers the T trading days of the $L \in \{1, \ldots, 5\}$ years before a formation date. The trading window E starts on the formation date and lasts $H \in \{1, 3, 6, 12\}$ months. A trading window starts every H months from January, so consecutive windows of one horizon overlap by at most 3 trading days. With normalized prices $\tilde{p}_{i,t} = P_{i,t}/P_{i,t_0}$, where t_0 is the last day of Φ, we define a spread of m stocks as

$$s_t = \sum_{i=1}^{m} w_i \, \tilde{p}_{i,t}, \qquad \|\mathbf{w}\|_1 = 1, \tag{1}$$

with weights estimated on Φ and held fixed over E. We compare 3 groupings of the eligible stocks. Both cluster groupings use daily returns from which the market mode is removed by a regression on the equal-weighted return of all eligible stocks [13].

Sectors. Each of the 11 sectors is one group.

OPTICS clusters. Within each sector, we represent each stock by its loadings on the PCs above the noise bound of random matrix theory [14], then cluster the stocks with ordering points to identify the clustering structure (OPTICS) [15], following Sarmento and Horta [5]. OPTICS does not require specifying the number of clusters and leaves stocks that fit no cluster unassigned. In the training sample, this gives about 24 clusters and leaves 43% of the stocks unassigned and excluded.

Complete-linkage clusters. Within each sector, the residual returns are clustered by complete linkage on a Spearman correlation distance. The number of clusters equals the number of PCs above the noise bound. In the training sample, this gives 31 to 45 clusters on average, depending on L.

3.2 Cointegrating Vectors

Table 1 lists the 5 methods, with the equations in Appendix A. Johansen, PCA and eigenanalysis weight all m stocks of a group jointly, and their spreads are called baskets.

Method	Weights	Formation statistic
Pairs	Regression of one normalized price on the other	Smaller ADF τ
Johansen	First cointegrating vector [3]	Trace statistic
PCA	Smallest PC of normalized price levels [4]	ADF τ of the spread
Eigenanalysis	Eigenvector of the smallest eigenvalue of the sum of squared autocovariances [16]	ADF τ of the spread
Stock versus group	Regression on the equal-weighted rest of its group	ADF τ of the residual

Table 1: Estimators of the cointegrating vector and their formation statistics, where ADF is the augmented Dickey–Fuller statistic τ and PCA is principal component analysis. All methods except pairs require a group of $m \geq 3$ stocks, and baskets require $T/m \geq 6$ [17].

3.3 Random Walk Null

Standard Dickey–Fuller critical values cannot be applied to the formation statistics, because each method chooses its weights to make the spread look stationary. For pairs, this is the residual-based testing problem, where critical values are simulated on independent random walks because the cointegrating vector is estimated [1, 18]. The distortion grows with the number of stocks. PCs of the levels of independent random walks yield residuals that a Dickey–Fuller test often finds stationary [19], and the Johansen test over-rejects when the system is large relative to the sample [17].

We therefore obtain the null distribution of each formation statistic by applying the method to log prices that follow independent driftless Brownian motions,

$$dX_{i,t} = \sigma_i \, dW_{i,t}, \qquad i = 1, \ldots, m, \tag{2}$$

where $W_{i,t}$ are independent standard Brownian motions, and each σ_i is drawn from the standard deviations of daily log returns from 2005 to 2019 of all stocks with at least 252 returns. Daily paths of length $T = 252L$ are normalized as $\tilde{p}_{i,t} = e^{X_{i,t} - X_{i,t_0}}$. For each method, we simulate 1,000 draws (or 10,000 for pairs).

The p-value p of a spread is the share of null statistics at least as extreme as its own. Spreads are assigned to deciles $d = \min\{\lfloor 10p \rfloor, 9\}$, so that $d = 0$ ($d = 9$) holds the smallest (largest) p-values. The proportion of true nulls among N spreads is estimated as $\hat{\pi}_0 = \min\{1, 2\,\#\{p \geq 0.5\}/N\}$ [20]. The simulated paths are extended over the trading window, and the null value of each measure defined below, for a spread in decile d, is its mean over the simulated spreads of the same method, m, T and H in decile d.

3.4 Trading Rule and Measures

Write $z_t = (s_t - \bar{s}_\Phi)/\hat{\sigma}_\Phi$, where $\bar{s}_\Phi$ and $\hat{\sigma}_\Phi$ are the sample mean and standard deviation of the spread over Φ. The position $n_t \in \{-1, 0, 1\}$ set at the close of day t follows the rule in [11],

$$n_t = \begin{cases} -\operatorname{sgn}(z_t) & \text{if } z_t \, n_{t-1} \geq 0 \text{ and } |z_t| \geq 2, \text{ and } t \text{ is not the last day of } E, \\ 0 & \text{if } n_{t-1} \neq 0, \; z_t \, n_{t-1} \geq 0 \text{ and } |z_t| < 2, \text{ or } t \text{ is the last day of } E, \\ n_{t-1} & \text{otherwise.} \end{cases} \tag{3}$$

A position opens when the spread is at least 2 standard deviations from its formation mean, and closes when the spread returns to the mean or the window ends. The net return r of a spread is its

gain over E on one unit of committed capital after trading and borrowing costs. Trading costs 10 basis points (bps) per unit traded, and borrowing costs 25 bps a year on the legs held short [21].

Persistence is measured by the ADF statistics τ_Φ and τ_E of the spread over the formation and trading windows, and by the rejection rate, the share of spreads whose τ_E rejects a unit root at the 5% level with the critical values of MacKinnon [18].

3.5 Portfolios and Inference

For each method, write $\bar{r}_{g,L,k}$ for the mean net return of the top-decile spreads of grouping g and window L in trading window k. With $\mathcal{C}_k$ the set of (g, L) that hold at least one top-decile spread in window k, the method portfolio return is the equal-weighted average

$$r_k = \frac{1}{|\mathcal{C}_k|} \sum_{(g,L)\in\mathcal{C}_k} \bar{r}_{g,L,k}. \tag{4}$$

Means of r_k over the K trading windows are annualized by multiplying by $12/H$, and t is the t-statistic of the mean with the heteroskedasticity and autocorrelation consistent variance of Newey and West [22], using $\lfloor 4(K/100)^{2/9} \rfloor$ lags. For $H = 6$ months, $K = 20$ in the training and 12 in the testing sample.

4 Results

4.1 Formation Statistics and Persistence

Formation p-values of pairs and stock-versus-group spreads are consistent with the null, with $\hat{\pi}_0$ between 0.95 and 1.00 (Table 2). For baskets, $\hat{\pi}_0$ is 0.66 to 0.77, so an estimated 23% to 34% of basket spreads depart from the random walk null at formation. Even under the null, top-decile baskets have τ_Φ between -7.2 and -8.1, far below the ADF critical values, because weights chosen over m independent random walks yield spuriously stationary spreads [19].

			τ_Φ		τ_E		Rejection rate (%)	
Method	$\hat{\pi}_0$	Spreads	Data	Null	Data	Null	Data	Null
Training, 2010 to 2019								
Pairs	1.00	6014	−3.77	−3.78	−1.57	−1.51	5.7	4.8
Johansen	0.70	84	−7.70	−7.17	−1.61	−1.51	5.9	4.8
PCA	0.66	84	−7.78	−7.64	−1.66	−1.53	7.2	5.2
Eigenanalysis	0.68	83	−7.90	−7.75	−1.65	−1.50	6.6	4.7
Stock versus group	0.95	592	−3.58	−3.57	−1.54	−1.51	5.0	5.1
Testing, 2020 to 2025								
Pairs	1.00	7579	−3.75	−3.78	−1.63	−1.51	5.5	4.8
Johansen	0.68	116	−8.25	−7.63	−1.69	−1.51	7.2	4.8
PCA	0.76	80	−7.80	−7.65	−1.75	−1.52	8.9	5.1
Eigenanalysis	0.77	84	−8.23	−8.07	−1.75	−1.49	9.0	4.7
Stock versus group	1.00	606	−3.56	−3.58	−1.62	−1.51	5.7	5.1

Table 2: Formation statistics and persistence, $H = 6$ months. Spreads is the mean number of top-decile spreads per trading window, and the τ and rejection-rate columns average the top-decile spreads as in (4). $\hat{\pi}_0$ is computed over all spreads.

Out of sample, the top-decile spreads behave almost like null spreads. Their τ_E lies within 0.26 of its null value of about -1.5, and the rejection rate is 5.0% to 5.7% for pairs and stock-versus-group spreads, against about 5% under the null, and 5.9% to 9.0% for baskets. Stock spreads are therefore at most weakly cointegrated, in line with Clegg [9] and Farago and Hjalmarsson [10].

4.2 Net Returns

No method is profitable (Table 3). Every top-decile portfolio loses money in both samples, and more in the testing sample, 0.70% to 1.34% a year against 0.28% to 0.97% in training, with $|t| > 3$ for every testing-sample loss. These losses agree with the near-zero or negative returns reported for recent pairs trading [6, 7, 8].

	Net return		Null	Net minus null		Top minus bottom	
Method	Mean	t	Mean	Mean	t	Mean	t
Training, 2010 to 2019							
Pairs	−0.97	−2.27	−1.12	0.15	0.35	0.27	0.43
Johansen	−0.28	−1.17	−1.07	0.79	3.13	1.67	3.56
PCA	−0.62	−3.58	−1.42	0.81	4.58	−0.40	−0.81
Eigenanalysis	−0.76	−3.24	−1.57	0.81	3.49	0.43	0.65
Stock versus group	−0.91	−3.20	−1.12	0.21	0.71	1.70	2.68
Testing, 2020 to 2025							
Pairs	−1.34	−3.07	−1.12	−0.23	−0.52	−0.14	−0.14
Johansen	−0.70	−3.89	−1.11	0.41	2.46	−0.07	−0.18
PCA	−0.84	−3.62	−1.46	0.61	2.39	0.20	0.38
Eigenanalysis	−0.78	−3.60	−1.70	0.91	3.51	0.59	0.86
Stock versus group	−1.06	−3.93	−1.14	0.07	0.27	2.07	1.48

Table 3: Annualized net return (%) of the top-decile portfolio (4), $H = 6$ months. Net minus null is the net return minus the null mean, and top minus bottom averages, as in (4), the difference between the top- and bottom-decile returns over the (g, L) that hold both.

Null spreads lose about their trading costs, and top-decile spreads lose less, except pairs in the testing sample. For pairs and stock-versus-group spreads, the net return minus the null is insignificant ($|t| < 1$), whereas baskets earn 0.4% to 0.9% a year more than the null ($t \geq 2.39$), in line with their higher rejection rates. Because the null has independent increments with constant variance, this excess return can also arise from common factors or volatility clustering, and it is not by itself evidence of cointegration. Formation p-values do not rank returns out of sample, as no top-minus-bottom return is significant in the testing sample.

The rule profits only from spreads that keep reverting over the trading window, and the top-decile rejection rates of 5.0% to 9.0% barely exceed the 5% of the null (Table 2), so cointegration-based formation does not make the spreads profitable. This agrees with Do and Faff [23], who attribute up to 70% of the decline in pairs trading profits to worsening arbitrage risks, which prevent or delay convergence. The losses accumulate steadily rather than in a few episodes (Figure 1). The first half of 2020, which contains the March 2020 crash, loses 0.41% on average across the 5 methods, close to the average half-year loss of the testing sample.

4.3 Windows and Grouping

All 20 combinations of formation window L and trading horizon H lose money in both samples (Table 4). The 1-month horizon loses most on average in both samples, and no formation window

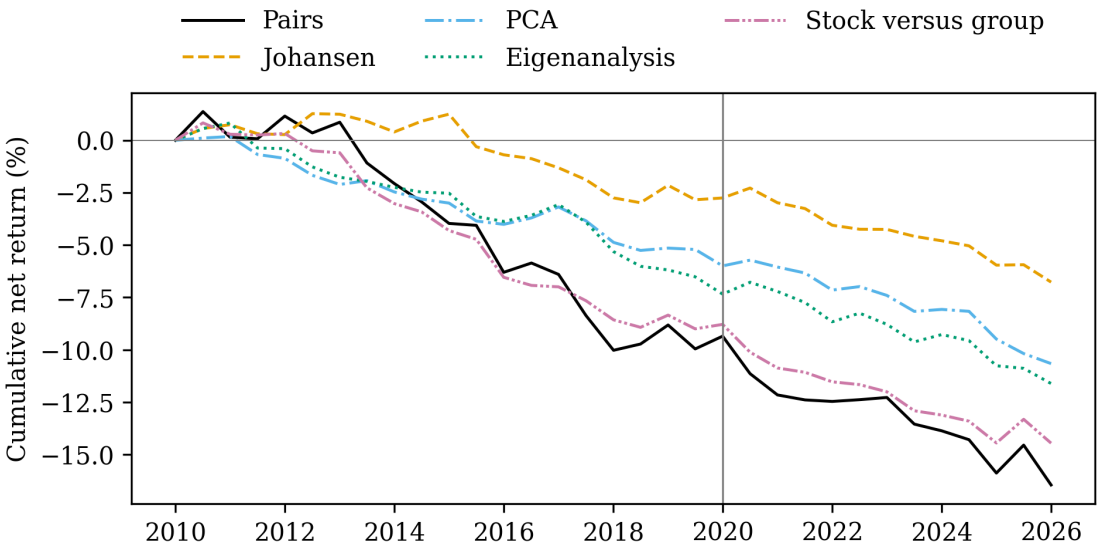


Figure 1: Cumulative net return of the top-decile portfolio (4) by method, $H = 6$ months. The vertical line marks the start of the testing sample.

is best across horizons or samples.

	Training, 2010 to 2019				Testing, 2020 to 2025			
L (years)	$H = 1$	$H = 3$	$H = 6$	$H = 12$	$H = 1$	$H = 3$	$H = 6$	$H = 12$
1	−1.79	−1.14	−1.12	−1.22	−1.49	−1.11	−0.48	−0.89
2	−1.15	−0.59	−0.54	−1.46	−1.49	−1.29	−0.92	−1.33
3	−1.08	−0.90	−0.92	−0.98	−1.79	−1.39	−1.57	−1.14
4	−1.03	−0.75	−0.55	−0.72	−1.07	−0.90	−1.04	−0.38
5	−1.21	−1.04	−0.67	−0.46	−1.12	−0.85	−0.82	−0.31

Table 4: Annualized net return (%) of the equal-weighted average of the 5 method portfolios over the 3 groupings, by formation window L and trading horizon H in months.

No grouping is profitable, but cluster spreads lose less than sector spreads in both samples. With $H = 6$ months, sectors lose 0.86% a year in training and 1.33% in testing, against 0.50% and 0.60% for OPTICS clusters and 0.73% and 0.98% for complete-linkage clusters. Out of sample, the 0.73% a year advantage of OPTICS clusters over sectors is significant, whereas the 0.35% advantage of complete-linkage clusters is not. These findings agree with evidence that cluster-based portfolios capture more than intra-industry mean reversion [24].

5 Conclusion

This study examines whether cointegration-based pairs of US large-cap stocks remain profitable out of sample, and whether their formation-period mean reversion persists, across 5 estimators of the cointegrating vector, sector and cluster groupings, 1- to 5-year formation windows and 1- to 12-month trading horizons, with independent random walks as the null.

Cointegration-based formation does not select profitable spreads. In the testing sample, the top-decile portfolio of every method loses 0.70% to 1.34% a year after trading and borrowing costs,

and the losses persist under the exit levels and costs of Appendix B. Neither the window nor the grouping changes their sign, and formation p-values do not consistently rank returns. Out-of-sample mean reversion exceeds that of random walks only marginally, in line with the worsening arbitrage risks that Do and Faff [23] link to the decline in pairs trading profits.

Two limitations qualify these results. First, Yahoo Finance holds no prices for stocks delisted before 2026, so the sample is subject to survivorship bias, which prices with delisting returns, such as those of CRSP, would remove. Second, the null of independent Brownian motions omits common factors and volatility clustering, so the 0.4% to 0.9% a year by which baskets beat the null is not by itself evidence of cointegration. A null with common factors and volatility clustering would separate the two sources and is left for future research.

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A Methodology Details

A.1 ADF Statistic

For a series s_t, the ADF regression with a constant and one lagged difference is

$$\Delta s_t = a + \gamma\, s_{t-1} + \phi\, \Delta s_{t-1} + \varepsilon_t, \tag{5}$$

where ε_t are the innovations, and $\tau = \hat{\gamma}/\operatorname{se}(\hat{\gamma})$. A more negative τ is stronger evidence that s_t is stationary.

A.2 Grouping

Market mode. With I_t the equal-weighted mean of the daily log returns of all eligible stocks, each stock's return is regressed over Φ as in [13],

$$\Delta \ln P_{i,t} = \mu_i + \kappa_i I_t + \eta_{i,t}. \tag{6}$$

Both cluster groupings use the residuals $\eta_{i,t}$ of the M stocks in a sector.

OPTICS clusters. Let C be the Pearson correlation matrix of the residuals. Eigenvalues of C above the upper edge of a random correlation spectrum [14],

$$\lambda_+ = \left(1 + \sqrt{M/T}\right)^2, \tag{7}$$

give J eigenvectors $\mathbf{e}_1, \ldots, \mathbf{e}_J$, and stock i is represented by its loadings $(e_{1,i}, \ldots, e_{J,i})$. OPTICS clusters these loadings with a minimum of 5 points. A sector with $J = 0$, or with fewer than 5 stocks, forms one group.

Complete-linkage clusters. Let ρ be the Spearman correlation matrix of the residuals. The tree of complete linkage on the distance δ_{ij} is cut into K clusters, where

$$\delta_{ij} = \sqrt{2(1 - \rho_{ij})}, \qquad K = \max\left\{1,\ \#\{\text{eigenvalues of } \rho > \lambda_+\}\right\}. \tag{8}$$

A.3 Cointegrating Vectors

Let $\tilde{\mathbf{p}}_t = (\tilde{p}_{1,t}, \ldots, \tilde{p}_{m,t})^\top$ be the normalized prices of a group over Φ with mean $\bar{\mathbf{p}}$. Each method gives a vector $\mathbf{v}$, normalized as

$$\mathbf{w} = \pm \frac{\mathbf{v}}{\|\mathbf{v}\|_1}, \tag{9}$$

with the sign chosen so that the largest weight in absolute value is positive.

Pairs. For stocks i and j,

$$\hat{\beta}_{ij} = \frac{\widehat{\mathrm{Cov}}(\tilde{p}_i, \tilde{p}_j)}{\widehat{\mathrm{Var}}(\tilde{p}_j)}, \qquad \mathbf{v} = (1, -\hat{\beta}_{ij}), \tag{10}$$

and likewise with i and j exchanged. The ordering with the smaller τ is kept.

Johansen. The vector error correction model

$$\Delta\tilde{\mathbf{p}}_t = \boldsymbol{\nu} + \Pi\tilde{\mathbf{p}}_{t-1} + \Gamma\Delta\tilde{\mathbf{p}}_{t-1} + \boldsymbol{\varepsilon}_t \tag{11}$$

where $\boldsymbol{\nu}$ is a constant, Π and Γ are $m \times m$ coefficient matrices and $\boldsymbol{\varepsilon}_t$ are the innovations, is estimated by reduced rank regression [3]. With $\mathbf{R}_{0,t}$ and $\mathbf{R}_{1,t}$ the residuals of $\Delta\tilde{\mathbf{p}}_t$ and $\tilde{\mathbf{p}}_{t-1}$ on $(1, \Delta\tilde{\mathbf{p}}_{t-1})$, and $S_{ab} = T^{-1}\sum_t \mathbf{R}_{a,t}\mathbf{R}_{b,t}^\top$ for $a, b \in \{0, 1\}$, the eigenvalues $\hat{\lambda}_1 \geq \cdots \geq \hat{\lambda}_m$ solve

$$\left|\lambda S_{11} - S_{10}S_{00}^{-1}S_{01}\right| = 0. \tag{12}$$

The vector $\mathbf{v}$ is the eigenvector of $\hat{\lambda}_1$, and the formation statistic is the trace statistic $\Lambda = -T\sum_{i=1}^{m}\ln(1 - \hat{\lambda}_i)$.

PCA. The vector $\mathbf{v}$ is the eigenvector of the smallest eigenvalue of the covariance of price levels [4],

$$\hat{\Sigma}_0 = \frac{1}{T}\sum_t (\tilde{\mathbf{p}}_t - \bar{\mathbf{p}})(\tilde{\mathbf{p}}_t - \bar{\mathbf{p}})^\top. \tag{13}$$

Eigenanalysis. The vector $\mathbf{v}$ is the eigenvector of the smallest eigenvalue of

$$\Omega = \sum_{h=0}^{5} \hat{\Sigma}_h \hat{\Sigma}_h^{\top}, \qquad \hat{\Sigma}_h = \frac{1}{T} \sum_{t} (\tilde{\mathbf{p}}_{t+h} - \bar{\mathbf{p}})(\tilde{\mathbf{p}}_t - \bar{\mathbf{p}})^{\top}, \tag{14}$$

with 5 lags as in the simulations of [16].

Stock versus group. For stock i, with $\tilde{p}_{-i,t}$ the equal-weighted mean of the normalized prices of the other $m-1$ stocks,

$$\hat{\beta}_i = \frac{\widehat{\mathrm{Cov}}(\tilde{p}_i, \tilde{p}_{-i})}{\widehat{\mathrm{Var}}(\tilde{p}_{-i})}, \qquad v_i = 1, \quad v_j = -\frac{\hat{\beta}_i}{m-1} \text{ for } j \neq i. \tag{15}$$

B Exit Levels and Costs

Table 5 reports the equal-weighted average of the 5 method portfolios, $H = 6$ months, with the exit condition $z_t\, n_{t-1} \geq 0$ of (3) replaced by $z_t\, n_{t-1} \geq -0.5$ and $z_t\, n_{t-1} \geq 0.5$, and with costs of 5 and 25 bps. Relative to the exit at the mean, the exit level changes the return by at most 0.17% a year, and every variant loses money in both samples.

		Training		Testing	
Exit at $\lvert z_t \rvert$	Cost (bps)	Mean	t	Mean	t
0.5 before the mean	5	−0.38	−2.02	−0.56	−3.76
	10	−0.76	−3.93	−0.96	−6.45
	25	−1.91	−8.80	−2.18	−12.64
0 at the mean	5	−0.36	−1.89	−0.58	−4.44
	10	−0.71	−3.62	−0.95	−7.04
	25	−1.75	−8.20	−2.04	−12.61
0.5 past the mean	5	−0.37	−1.89	−0.55	−4.53
	10	−0.69	−3.47	−0.89	−7.09
	25	−1.66	−7.77	−1.91	−12.58

Table 5: Annualized net return (%) of the equal-weighted average of the 5 method portfolios under alternative exit levels and costs, $H = 6$ months.